

CSCE 190: Computing in the Modern World

Class 4 – Resources (HTML/CSS, Markdown) via GitHub

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Carolinian Creed: “I will practice personal and academic integrity.”

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Outline

- Introduction Section
 - Recap from Last Class
- Main Section
 - Sharing Resources – Content Format
 - HTML, CSS
 - Markdown
 - GitHub pages
- Conclusion Section
 - Ask Me Anything

Introduction Section

Recap from Last Lecture - Class 3 + Lab

We talked about

- Sharing Resources – Importance and What to Watch Out For
- GitHub
- GitHub pages
- Touched upon HTML and Markdown

Activity 3-1 – GitHub

- Objective: learn your way through Git and Github
- Tasks
 - Create an account on GitHub
 - Access GitHub repo created in class:
 - [https://github.com/AI4S-AI-teaching-resources//fall2026-class190-section\[X \]-demo1](https://github.com/AI4S-AI-teaching-resources//fall2026-class190-section[X]-demo1) where X is your section.
Example
<https://github.com/AI4S-AI-teaching-resources/fall2026-class190-section4-demo1>
 - See contents
 - Now create your own repo under your account
 - Call it CSCE-190-[yourname], keep it private
 - Create a sub-folder called **assignments**
 - Put your mission statement (from Class 2) under **assignments**.

Activity 3-2 – GitHub

- Objective: learn how to collaborate on a file
- Tasks
 - Create a public repo called '**test-content-collaboration**'
 - Create a file called '**event-calendar.md**'
 - Create an entry of an event: name, place, time
 - Ask your two neighbor to edit add an event to the calendar
 - They will have to fork
 - Checkout the calendar file
 - Make change
 - Add/ Commit/ Push changes
 - You will have to merge the change

Activity - Game: CyberSecurity

- DART Academy - <https://dartacademy.net/about>
- DeepCover - <https://dartcollective.net/deepcover/>

Main Section

Editor Concept: Format and Content

WYSIWYG – what you see is what you get

- Software that lets you edit content so it looks just like the final result (format)
- Pros
 - Easy to edit
- Cons
 - If format changes slightly, may spend a lot of time fixing it, even if the content is same

Examples

- Word Processors: Programs like Microsoft Word or Google Docs.
- Website Builders: Tools like Wix or WordPress where you drag and drop items.
- Email Clients: Gmail or Outlook when you change text color or add links

Editor Concept: Format and Content

Typesetting – separate content from format

- Software that lets you edit content separately;
- Format can be changed separately as well
- Pros
 - Easy to convert format
 - Advanced styles can be used seamlessly
- Cons
 - Harder to use software

Examples

- Adobe InDesign: The professional industry standard for books, magazines, and complex layouts.
- LaTeX: A code-based system ideal for heavy academic, mathematical, and technical documents.
- HTML/ CSS: html for content, css for digital styling

Concept: HTML/ CSS

HTML – Hypertext Markup Language

- DOCTYPE
- HTML
- HEAD
- TITLE
- BODY
- H1
- P

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Credit: https://www.w3schools.com/html/html_basic.asp

Exercise: HTML - Elements

HTML – Add HEAD and TITLE tags to the example

- DOCTYPE
- HTML
- **HEAD**
- **TITLE**
- BODY
- H1
- P

```
<!DOCTYPE html>
<html>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Credit: https://www.w3schools.com/html/html_basic.asp

Concept: HTML - Link

HTML – Link

```
<a href="url">link text</a>
```

```
<a href="https://www.w3schools.com/">  
Visit W3Schools.com!</a>
```

Credit and exercise:

https://www.w3schools.com/html/tryit.asp?filename=tryhtml_links_w3schools

Concept: HTML - List

HTML – List

```
<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>
```

1. Coffee
2. Tea
3. Milk

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

- Coffee
- Tea
- Milk

Credit and exercise:

https://www.w3schools.com/html/html_lists.asp

Concept: HTML - Table

HTML – Table

```
<table>
<tr>
  <th>Company</th>
  <th>Contact</th>
  <th>Country</th>
</tr>
<tr>
  <td>Alfreds Futterkiste</td>
  <td>Maria Anders</td>
  <td>Germany</td>
</tr>
<tr>
  <td>Centro comercial Moctezuma</td>
  <td>Francisco Chang</td>
  <td>Mexico</td>
</tr>
</table>
```

Company	Contact	Country
Alfreds Futterkiste	Maria Anders	Germany
Centro comercial Moctezuma	Francisco Chang	Mexico

Credit and Exercise link:

https://www.w3schools.com/html/tryit.asp?filename=tryhtml_table3

HTML Cheat Sheet, Standard

- Categories: https://www.w3schools.com/Tags/ref_byfunc.asp
 - Basic
 - Formatting
 - Forms and input
 - Frames
 - Images
 - Audio/ video
 - **Links**
 - **Lists**
 - **Tables**
 - Styles and semantics
 - Meta info
 - Programming

HTML Cheat sheet:

<https://web.stanford.edu/group/csp/cs21/htmlcheatsheet.pdf>

Authoritative source: <https://html.spec.whatwg.org/multipage/>

Concept: CSS

- Defines core CSS properties for controlling text presentation fall into a few primary categories:
 - **font-family**: Specifies the typeface. It accepts a fallback list of fonts ending in a generic family (e.g., serif, sans-serif, monospace).
 - **font-size**: Sets the size of the text.
 - **line-height**: Controls the vertical spacing between lines of text (known as *leading* in print).
 - **font-weight**: Alters the boldness using numeric values (e.g., 400 for normal, 700 for bold).
 - **letter-spacing**: Manages the spacing between characters (*tracking*)

HTML / CSS Resources

- W3Schools: <https://www.w3schools.com/html/>
- Codecademy: <https://www.codecademy.com/learn/learn-html>
- Mozilla tutorials: <https://developer.mozilla.org/en-US/docs/MDN/Tutorials>
- Freecodecamp videos: <https://youtube.com/c/Freecodecamp>
- Google web.dev: <https://web.dev/learn/html>
- Udemy (filtered by free and “HTML”): https://www.udemy.com/courses/development/web-development/?course_label=6148&price=price-free&sort=most-reviewed

Credit: Robert Bailey

Document using Markdown

- Background
 - A simple markup language for the web
- Resources
 - Basics: <https://www.markdownguide.org/basic-syntax/>
 - Cheat sheet: <https://www.markdownguide.org/cheat-sheet/>

Activity 4 – HTML / CSS, Markdown

- Objective A: learn HTML/ CSS
- Tasks
 - See HTML cheat sheet
 - Create HTML files with an editor (like Notepad, vi) or online tool
 - Learn about different constructs
- Objective B: learn Markdown
- Tasks
 - See Markdown cheat sheet
 - Create Markdown files with an editor (like Notepad, vi)
 - Learn about different constructs

Concluding Section

Conclusion

We talked about

- Sharing Resources – Content Format
- HTML, CSS
- Markdown
- GitHub pages

Week	Content	Assignment
1	Intro, syllabus, university resources	Slack account
2	CSE majors, career paths, research areas	Write a mission statement
3	Github (account and basics) + Lab	Github account, repo
4	HTML basics, IDE (VS Code)	None
5	More HTML, CSS, Problem Solving	Update Github with CSS
6	Brainstorming, problem statement	Problem statement
7	Computing – (Software) Dev. Life Cycle	Draft Requirement, Specification
8	Computing – Testing a System	Test Plan
9	AI – LLMs/FMs, Generative AI, Issues and Challenges	Problem Solving with GenAI
10	Building a resume, LinkedIn	Draft resume in PDF
11	Review resumes	Review resumes, update Github
12	Convert resume to HTML	Begin conversion, update Github
13	CSE research areas	(Finalize website)
N/A	Last day of classes: 4 Dec, 2026	All extensions end

Ask Me Anything
